



Precision Industries
Testing Department

GENERATING-SET

LOAD TEST

Model No: PI 250 C

60 Hz

Type: Close

Job No: 9258

Color: Orange

Date: 08/06/2018

Serial Number:								
PI Model No:	PI 250 C	Prime Power:	225	[kVA]	P.F (Load):	1	Voltage:	380

Generating Set Details:

Engine:	Cummins	Model No: QSB7 - G5	Serial No: 84463920
Alternator:	Stamford	Model No: UCI274H1	Serial No: N18C104029
Controller:	DSE	Model No: 6120	Serial No: 6754497
C. Breaker:	Metasol 630 A	Model No: ABN 803 C	Serial No: N/A
C.Transformer:	Ziegler 400/5A	Model No: N/A	Serial No: N/A
AVR:	Stamford	Model No: SX 460	Serial No: 112943 - 0323
Cooling System: Water Cool	Enclosure Type: Canopy	Protection Class: IP23	

Load Test Ambient Conditions:

Temperature	40	[°C]	Altitude:	0	[m]	Humidity:	63	%
-------------	----	------	-----------	---	-----	-----------	----	---

Generating Set Rating:

Testing Full Load Current:	273.48	Amps	Nominal Current:	341.9	Amps
Power Correction Factor (De-rating):	0.000	%	Corrected Prime Power:	0.0	[kVA]

LOAD TEST

Load	Run Time	Freq	Speed	VOLTAGE L-L			CURRENT			Power	Coolant Temp	Oil Pressure	Charg-Alt.
				L1	L2	L3	L1	L2	L3				
%	min	Hz	RPM	V	V	V	A	A	A	kW	C	bar	VDC
0 %	5	60	1800	380	380	380	-	-	-	-	71	4.8	14.6
25 %	5	60	1800	380	380	380	65	64	64	42.0	82	4.35	14.6
50 %	5	60	1800	380	380	381	136	135	135	89.0	84	4.16	14.6
75 %	5	60	1800	380	381	381	206	205	204	135.0	86	3.96	14.6
100 %	10	60	1800	381	381	382	276	275	275	182.0	87	3.78	14.6
110 %	10	60	1800	381	382	382	300	300	300	197.0	89	3.57	14.6

LOAD ACCEPTANCE TEST

Suddenly Block Load %	Voltage [V]	Frequency [Hz]	Current [A]	Response Time [sec]
50%	380	60.0	136	0
0%	385 - 380	62.7 - 60.0	0	2
50%	374 - 380	58.8 - 60.0	140	2

PROTECTION SYSTEM TEST

Over Frequency SD	Under Frequency SD	Over Voltage SD	Under Voltage SD	Oil Pressure SD	Coolant Level SD	Coolant Temp SD
65 / 67	55 / 53	440 / 460	360 / 340	1.0	OK	103 / 108 c

ACCESSORIES DETAILS

Model:	Serial No.
Model:	Serial No.

TESTED
Testing Department

PI Document no.

PI-TST-LT-01

Rev.

01

Test Certificate

FOR
SALIENT POLE, SELF EXCITED / PMG EXCITED AND SELF REGULATED AC GENERATORMachine No. **N18C104029**

Type Test has been conducted on similar design machine
STANDARDS: Generator generally conforms to BS 5000 : Part 3,
 Tested in accordance with BS EN 60034-1/IS/IEC 60034-1
 Rotor dynamically balanced to BS ISO 1940 - 1

ROUTINE TEST CERTIFICATE FOR BRUSHLESS AC GENERATOR:

FRAME: UCI274H1	AVR: SX460	AVR NO: 1129430323	ENCLOSURE: IP23	
RATED KVA: 220	RATED VOLTS: 400	RATED AMPS: 317.5	FREQUENCY: 50	PHASE: 3
POWER FACTOR: .8	STR CONN: Series Star	STR WINDING: 311	DUTY: Peak Stand-by	WIRE: 4-Wire
EXC. VOLTS: 58	EXC.AMPS: 2.3	AMB TEMP: 27		

FOR 3 PHASE SEQUENCE LOOKING FROM DRIVE END FOR CLOCKWISE ROTATION :U-V-W

WINDING	COLD RESISTANCE IN Ohm(+/- 10 Milli Ohm) AT 20°C (MAX VALUES)	INSULATION RESISTANCE (M- Ohm)	HV TEST 2KV FOR 1 MIN	INSULATION CLASS
STATOR	0.03371	2+	OK	H
ROTOR	1.9663	2+	OK	H

REGULATION TEST

FREQ.	VOLTS	AMPS	KVA	PF	KW	EXC.VOLT	EXC.AMP	% LOAD
52	401.35	0.00	0.00	0.00	0.00	12	.5	0.00
50	400	317.5	220	.8	176	58	2.3	100.00

REGULATION = (N.L.VOLTAGE-F.L.VOLTAGE / RATED VOLTAGE)*100 = 0.34 %

Item Description: AC Generator.UC27.4.H.1.311.,2,,11.5.,SX460.Black - RAL9011
 Remarks:

Date: 09-MAR-18	Job No: 3510145-003	AUTHORISED BY Mahesh R Tarde Quality Assurance
-----------------	---------------------	---

Certificate Of Conformity

Quality System Approval
 ISO 9001/2008

Frame Size : **UCI274H1**Serial Number : **N18C104029**

Manufactured Date : 09-MAR-18

KVA: **220**Authorized By Quality Assurance: **Mahesh R Tarde**